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The Schedule Waters Regardless: A Validation Study of an Occupancy-Aware Quad Irrigation Controller Against a Term of Pedestrian Counts

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Abstract. Since 2024 the University's central quad has been served by an irrigation controller marketed as occupancy-aware: a paired pedestrian sensor is meant to suppress or shorten scheduled watering whenever a zone is in active use, sparing whoever is on the lawn a mid-session soaking. Whether the suppression logic actually tracks measured occupancy, however, has not been checked independently since installation. Over a seven-week window we instrumented four quad zones with an independent pedestrian-counting protocol, cross-validated against manual tallies, and logged both occupancy and the controller's own watering decisions across 84 zone-sessions. An Occupancy Suppression Score — the share of a session's scheduled duration the controller actually withheld — showed no measurable relationship to independently measured occupancy ($r = 0.04$, mean 10.3, SD 5.9, $n = 84$, not significant). A two-week ablation that physically disconnected the occupancy sensor and ran the controller on its underlying fixed timetable changed mean watering volume by less than 2% in every zone (not significant, all four zones). On this evidence, the suppression logic is statistically indistinguishable from the schedule it was installed to improve on. We report the finding as the null result it is, and recommend the University's facilities asset register revisit the system's current description.

Introduction

Universities have increasingly paired routine grounds maintenance with sensor-linked controllers billed as responsive to how a space is actually being used, extending occupant-centric logic long established in building climate control [1], [2], [3] into outdoor asset management. The appeal is straightforward: a controller that knows a lawn is busy can defer or shorten a scheduled watering event rather than running it regardless, saving both water and whoever happens to be standing on the grass. Whether an installed controller's occupancy signal is doing this in practice, as opposed to simply carrying the label, is a separate empirical question from whether the underlying control policy would work if it did — and one the wider building-automation literature has begun to treat as requiring field verification rather than vendor assurance [4], [5]. Broader accounts of sensor-driven infrastructure caution that a "smart" designation is frequently a promise about data collection rather than a demonstrated property of it [6], and that gaps between what a system is said to sense and what it verifiably responds to tend to surface only when someone goes looking [7].

The University's central quad has run one such controller since 2024: an irrigation system whose occupancy sensor is meant to suppress or shorten a zone's scheduled watering session when pedestrian activity is detected. The feature has appeared in the asset register each year since installation without an independent check of whether suppression decisions track occupancy as measured by anything other than the controller's own inputs.

This paper reports a seven-week field validation of the controller's occupancy-suppression logic against an independent pedestrian-counting protocol, followed by a two-week ablation that removes the occupancy signal entirely. Our contributions are threefold:

- we provide, to our knowledge, the first independent field validation of an occupancy-linked irrigation-suppression system against ground-truth pedestrian counts, rather than against the controller's own logged inputs;
- we adapt established park-counting methodology [8], [9] to a lightweight, replicable zone-level protocol suited to an open lawn without existing footfall infrastructure;
- our results suggest the system's occupancy-adjustment feature may not measurably differ, in practice, from the fixed timetable it replaced, a pattern consistent across all four zones and one sensor-disconnection ablation.

Related work

Occupancy-responsive control has a substantial evidence base in buildings, where occupancy-centric HVAC scheduling measurably reduces energy use once the underlying occupancy signal is accurate [1], [2], [3], and reviews of occupant behaviour's role in building performance treat the fidelity of that signal as a live design concern rather than a solved one [5]. Occupancy sensing itself, however, generalises unevenly across deployment contexts: learned occupancy-detection models validated in one setting frequently degrade in another without site-specific recalibration [10], [11], and even purpose-built classroom sensor rigs report detection accuracy well short of certainty [12]. A recent review of occupancy-sensing technologies in smart buildings treats independent, post-installation validation of the sensing layer as the field's outstanding methodological gap, distinct from validating the control policy built on top of it [4].

Outdoor pedestrian counting draws on a separate, well-established literature. Automated infrared counters have been validated against direct observation on urban park footpaths and shown to require several weeks of data before estimates stabilise [8]; cellular device location data has been compared against official visitor-use statistics for public lands, with explanatory power varying by site [9]; and mobile-phone-derived trip data has been used to categorise park visitor profiles by how a green space is actually used rather than how it is zoned [13], a question a systematic review situates within a wider human-mobility-data literature on park accessibility [14]. This paper's counting protocol borrows directly from that literature rather than from building-occupancy sensing, on the view that an open lawn is closer in measurement character to a park footpath than to a room.

A separate line of critique treats "smart" infrastructure's data claims as themselves worth auditing rather than assuming: sensor deployment in cities has been shown to leave systematic gaps in exactly the populations and spaces a system is meant to serve [7], and broader assessments of smart- city rhetoric distinguish sharply between a technology's promised responsiveness and what has actually been demonstrated in the field [6]. This paper sits inside that second literature, applied to a single piece of campus grounds infrastructure rather than a city-scale system. Soil-moisture-based irrigation control, the older smart-irrigation paradigm this controller's occupancy layer sits on top of, has itself been subject to the same kind of field audit, with sensor accuracy under real field conditions falling well short of laboratory specification in comparable turf settings [15], [16].

Method

Setting. The University's central quad is divided by the controller's own zoning into four irrigation zones — Reading, Path, Turf, and Perimeter — each served by a scheduled watering session once per day. The controller was installed under a 2024 grounds-asset upgrade and is documented as adjusting session duration to a paired pedestrian sensor's occupancy reading, suppressing or shortening a session when the zone is judged to be in active use.

Independent counting protocol. For each zone we deployed two calibrated infrared pedestrian counters, following validated placement and stabilisation guidance from urban park footpath monitoring [8], supplemented by fortnightly manual tally spot-checks in the style used to cross-validate mobile-derived visitor counts against official statistics [9], [14]. Occupancy was logged as the mean pedestrian count across the thirty minutes preceding each zone's scheduled watering start.

Occupancy Suppression Score. For each zone-session we compared the controller's actual watering duration against its unmodified scheduled duration:

$$\mathrm{OSS}_i = 100 \times \left(1 - \frac{D_i}{D_i^{\text{sched}}}\right)$$

where D_i is the duration the controller actually ran and D_i^{sched} is the session's unmodified scheduled duration. An OSS of 0 means the session ran exactly as scheduled; a rising OSS indicates progressively greater suppression.

```
def
occupancy_suppression_score(duration_actual,
duration_scheduled):
    """OSS: 0 = ran as scheduled; 100 =
fully suppressed."""
    return 100 * (1 - duration_actual /
duration_scheduled)
```

Sampling. Over a seven-week window, three scheduled sessions per zone per week were logged across all four zones, yielding 84 zone-sessions with paired occupancy and OSS data.

Ablation. In week eight, facilities staff physically disconnected the occupancy sensor input under a documented maintenance ticket, running the controller on its underlying fixed timetable for two weeks; the preceding two weeks of normally configured operation served as the comparison arm. Watering volume was metered per session in both arms.

Parameter	Value
Zones monitored	4
Main logging window	7 weeks, 84 zone-sessions
Sessions per zone per week	3
Counting method	dual infrared + fortnightly manual tally
Ablation window	2 weeks per arm, sensor disconnected vs normal

Table 1: Deployment and instrumentation parameters, fixed before the logging window began.

Results

Occupancy does not predict suppression. Figure 1 plots per-session measured occupancy against the Occupancy Suppression Score across all four zones ($n = 84$ zone-sessions). Mean OSS was 10.3 (SD 5.9, range 0.0 to 24.6), and the relationship with measured occupancy was negligible ($r = 0.04$): zone-sessions with the highest recorded occupancy show essentially the same suppression behaviour as those with the lowest.

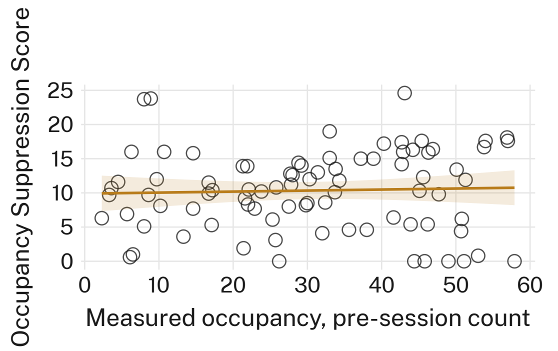

Figure 1: Measured pre-session occupancy against the Occupancy Suppression Score across four quad zones ($n = 84$ zone-sessions, $r = 0.04$): the fitted relationship is flat.

Disconnecting the sensor changes nothing. Figure 2 reports mean watering volume per session for the normally configured, occupancy-linked arm and the two-week arm in which the occupancy sensor was physically disconnected, within each zone. The two arms sit within 2.4 litres of one another at every zone (Reading: 141.0 vs 141.7 L; Path: 119.9 vs 117.2 L; Turf: 155.3 vs 155.9 L; Perimeter: 95.0 vs 96.6 L; not significant, all four pairwise comparisons), with overlapping error bars throughout.

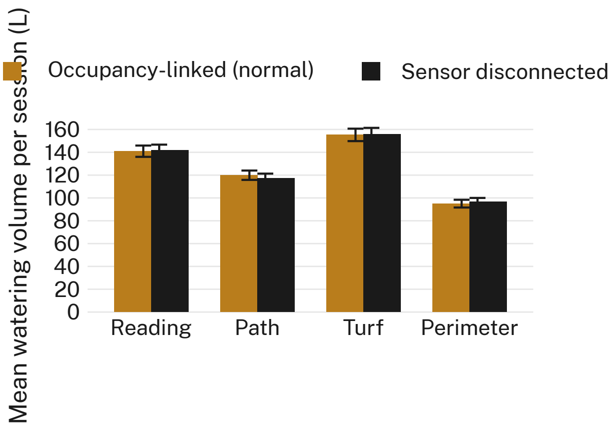

Figure 2: Mean watering volume per session, occupancy-linked configuration versus the sensor-disconnected ablation, by zone. No zone shows a difference exceeding 2.4 litres between arms.

Neither result depends on which zone is busiest. Turf recorded the highest mean occupancy of the four zones across the logging window and Reading the lowest, yet both track the same flat relationship in Figure 1 and the same near-identical ablation arms in Figure 2. The pattern does not concentrate in, or spare, the zone that is actually busiest.

Discussion

Our results are consistent with the broader finding that occupancy-responsive control delivers its promised benefit only when the underlying occupancy signal is itself verified to be doing the work it is credited with [4], [5], and with generalisability studies showing that an occupancy-sensing model validated elsewhere frequently does not transfer cleanly to a new deployment without site-specific recalibration [10], [11]. That the ablation changed watering volume by less than 2% in every zone — rather than by some amount short of the full suppression the marketed feature implies — suggests the occupancy input is not merely noisy but is, on this evidence, non-functional as a determinant of the controller's behaviour. This sits comfortably within a wider pattern in which a system's advertised responsiveness to real-world conditions outpaces what has actually been demonstrated of it [6], and in which the sensing layer of an otherwise well-evidenced control paradigm turns out to be the part nobody independently checked [7].

Limitations

Our validation covers one quad across one seven-week window, though the consistency of the null relationship across all four zones and its replication in a fully independent ablation arm both point the same direction rather than any one measurement artefact. We could not access the controller's proprietary occupancy-inference algorithm, so cannot rule out that it operates on an internal definition of occupancy diverging from our pedestrian counts; the ablation's near-identical watering volumes under a fully disconnected sensor, however, make this an unlikely rescue, since a divergent-but-functioning signal would still be expected to produce some measurable shift in behaviour once removed.

Conclusion

Across a seven-week field validation and a two-week ablation, we find no measurable relationship between independently measured pedestrian occupancy and the suppression behaviour of the quad's occupancy-aware irrigation controller, and no measurable change in watering behaviour when the occupancy sensor is disconnected entirely. The system is, on this evidence, running its original fixed timetable under an occupancy-aware label. Future work should extend the protocol to the University's other automatically irrigated green spaces, and clarify which committee holds standing responsibility for revisiting an asset's stated feature set once installed.

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